

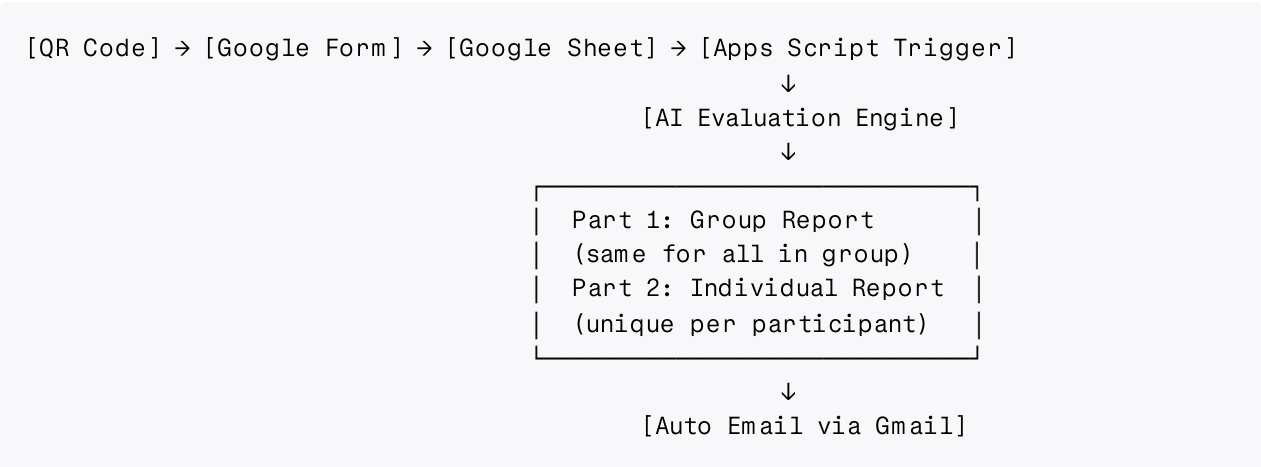
AI Literacy Survey — End-to-End Automated Workflow Framework

For SIA AI Literacy Assessment (Architects)

Executive Summary

This framework documents the complete, automated pipeline for the SIA AI Literacy Survey — from QR code scan to personalised AI-generated feedback delivery. The workflow uses Google Forms → Google Sheets → Apps Script → AI (ChatGPT/Gemini API) → Gmail, with zero manual intervention after initial setup. Every participant receives a group-level competency report (Part 1, common) and a personalised individual assessment (Part 2, unique).[^1]

Section 1: Survey Architecture Overview



The entire pipeline is serverless — hosted within Google Workspace — with the AI evaluation layer called via a REST API (OpenAI or Google Gemini). No third-party middleware is required unless preferred (e.g., Zapier or [Make.com](#) as alternatives).

Section 2: Google Form Setup

2.1 Form Fields (in order)

#	Field	Type	Required
1	Name of Firm / Group	Short text	Yes
2	Your Full Name	Short text	Yes

#	Field	Type	Required
3	Email Address	Email validation	Yes
4	Years in Practice	Multiple choice: < 2 yrs / 2-5 yrs / 6-10 yrs / 11-20 yrs / > 20 yrs	Yes
5	Mobile Number	Short text	No (optional)
6–15	MCQ Questions Q1–Q10	Multiple choice (A/B/C/D)	Yes

Form Settings to enable:

- ☒ Collect email addresses (for auto-reply)
- ☒ Limit to 1 response (optional, to prevent duplicates)
- ☒ Confirmation message: *"Thank you! Your personalised AI Literacy Report will be emailed to you shortly."*
- ☒ Do NOT enable "Edit after submit" — this prevents re-scoring

2.2 QR Code Generation

Once the Google Form is published, copy the shareable link. Generate a QR code using any free tool (e.g., [qr-code-generator.com](https://www.qr-code-generator.com/) or Google's own QR API). The QR code can be printed, projected on screen, or embedded in event materials. The form opens natively in mobile browsers — no app download required.[^1]

Section 3: The 10 MCQs — With Answer Key

[^1]

The questions are drawn from the uploaded SIA AI Literacy Survey MCQ set and are structured across five competency domains.

Q#	Category	Question Summary	Correct Answer
Q1	Foundational	What does 'generative AI' best describe?	B — AI that creates new content (text, images, designs) based on a prompt
Q2	Foundational	Best example of AI already in architect's tools?	B — Automated clash detection in BIM platforms
Q3	Applied	What is 'prompt engineering'?	C — Crafting clear instructions to guide AI output
Q4	Applied	Best AI tool for exploring 3D massing configurations?	B — Generative design / AI-assisted parametric tools
Q5	Critical Thinking	What is 'AI hallucination'?	B — AI confidently producing incorrect or fabricated information

Q#	Category	Question Summary	Correct Answer
Q6	Ethics / Legal	Who retains professional responsibility for AI-assisted design?	D — The architect of record
Q7	Technical	How do LLMs (e.g., ChatGPT, Claude) generate responses?	C — Predicting statistically likely next word/phrase from training data
Q8	Critical Thinking	Biggest professional risk when using AI for specification writing?	B — AI may generate outdated, inaccurate, or jurisdiction-specific clauses
Q9	Applied	How do AI image tools handle architectural spatial requirements?	B — Visually convincing but no understanding of spatial logic or codes
Q10	Self-Assessment	Current level of AI tool use in practice?	No wrong answer — used to segment cohorts

Note on Q10: Q10 is a self-assessment question with no correct answer. It is excluded from the score calculation (max score = 9 from Q1–Q9) but used to enrich the AI-generated narrative.

3.1 Scoring Rubric and Competency Tiers

Score (out of 9)	Competency Tier	Recommended Learning Path
0–3	☒ Foundational Awareness	Foundational AI Awareness Module
4–6	☒ Applied Practitioner	Applied AI Tools Workshop
7–8	☒ Strategic & Critical	AI Strategy & Ethics Masterclass
9	☒ AI Champion	Peer-Led AI Practice Group

Section 4: Google Sheets Setup

4.1 Automatic Response Sheet

When Google Forms is linked to Sheets (via *Responses* → *Link to Sheets*), a new sheet named "Form Responses 1" is auto-created with columns matching the form fields. Columns will auto-populate on each submission.^[1]

Expected columns (auto-generated):

Column	Header
A	Timestamp
B	Name of Firm / Group
C	Your Full Name
D	Email Address
E	Years in Practice

Column	Header
F	Mobile Number
G	Q1 Answer
H	Q2 Answer
I	Q3 Answer
J	Q4 Answer
K	Q5 Answer
L	Q6 Answer
M	Q7 Answer
N	Q8 Answer
O	Q9 Answer
P	Q10 Answer (Self-Assessment)

4.2 Scoring Sheet (add manually — "Scoring" tab)

Create a second sheet tab called "Scoring" with these computed columns, using formulas referencing the response sheet:

Column	Header	Formula (example for row 2)
A	Participant Name	= 'Form Responses 1' !C2
B	Group	= 'Form Responses 1' !B2
C	Email	= 'Form Responses 1' !D2
D	Years in Practice	= 'Form Responses 1' !E2
E	Q1 Score	= IF ('Form Responses 1' !G2="B" , 1, 0)
F	Q2 Score	= IF ('Form Responses 1' !H2="B" , 1, 0)
G	Q3 Score	= IF ('Form Responses 1' !I2="C" , 1, 0)
H	Q4 Score	= IF ('Form Responses 1' !J2="B" , 1, 0)
I	Q5 Score	= IF ('Form Responses 1' !K2="B" , 1, 0)
J	Q6 Score	= IF ('Form Responses 1' !L2="D" , 1, 0)
K	Q7 Score	= IF ('Form Responses 1' !M2="C" , 1, 0)
L	Q8 Score	= IF ('Form Responses 1' !N2="B" , 1, 0)
M	Q9 Score	= IF ('Form Responses 1' !O2="B" , 1, 0)
N	Q10 Response	= 'Form Responses 1' !P2

Column	Header	Formula (example for row 2)
O	Total Score (Q1-Q9)	=SUM(E2:M2)
P	Competency Tier	=IFS(O2<=3,"Foundational",O2<=6,"Applied Practitioner",O2<=8,"Strategic & Critical",O2=9,"AI Champion")

Section 5: Google Apps Script — Automation Engine

5.1 How It Works

Google Apps Script (GAS) is a built-in JavaScript runtime within Google Workspace. A **form submit trigger** fires automatically each time a participant submits. The script:[^1]

1. Reads the new response row from the Scoring sheet
2. Computes the participant's score and tier
3. Queries all rows with the same Group name → computes group average and tier distribution
4. Calls the AI API (OpenAI GPT-4 or Google Gemini) with a structured prompt
5. Parses the AI response into Part 1 (group) and Part 2 (individual)
6. Sends a formatted email to the participant via Gmail

5.2 Complete Apps Script Code

```
// =====
// AI LITERACY SURVEY – AUTO-EVALUATION & EMAIL SCRIPT
// Paste in Google Apps Script Editor (Extensions > Apps Script)
// =====

const SHEET_NAME = "Scoring";
const OPENAI_API_KEY = "sk-YOUR_KEY_HERE"; // Replace with your API key
const FROM_NAME = "SIA AI Literacy Survey";

// Answer key (correct options for Q1-Q9)
const ANSWER_KEY = { Q1:"B", Q2:"B", Q3:"C", Q4:"B", Q5:"B", Q6:"D", Q7:"C", Q8:"B", Q9:"C" };

// — TRIGGER FUNCTION (set as onFormSubmit trigger) —
function onFormSubmit(e) {
  const ss = SpreadsheetApp.getActiveSpreadsheet();
  const sheet = ss.getSheetByName(SHEET_NAME);
  const data = sheet.getDataRange().getValues();
  const headers = data[0];
  const lastRow = data[data.length - 1]; // most recent submission

  // Extract participant info
  const participantName = lastRow[headers.indexOf("Participant Name")];
  const group = lastRow[headers.indexOf("Group")];
```

```

const email          = lastRow[headers.indexOf("Email")];
const yearsInPractice = lastRow[headers.indexOf("Years in Practice")];
const totalScore     = lastRow[headers.indexOf("Total Score (Q1-Q9)")];
const tier            = lastRow[headers.indexOf("Competency Tier")];
const q10            = lastRow[headers.indexOf("Q10 Response")];

// Get individual question scores (columns E-M = index 4-12)
const qScores = [];
for (let i = 4; i <= 12; i++) qScores.push(lastRow[i]);

// — Compute Group Stats —
const groupRows = data.slice(1).filter(row => row[headers.indexOf("Group")] !== "");
const groupScores = groupRows.map(row => row[headers.indexOf("Total Score (Q1-Q9)"]);
const groupAvg = groupScores.reduce((a,b) => a+b, 0) / groupScores.length;
const tierCounts = { "Foundational":0, "Applied Practitioner":0, "Strategic & Critical":0 };
groupRows.forEach(row => {
  const t = row[headers.indexOf("Competency Tier")];
  if (tierCounts[t] !== undefined) tierCounts[t]++;
});
const groupSize = groupRows.length;

// — Build AI Prompt —
const prompt = buildPrompt({
  participantName, group, email, yearsInPractice,
  totalScore, tier, q10, qScores, groupScores,
  groupAvg, tierCounts, groupSize
});

// — Call OpenAI API —
const aiResponse = callOpenAI(prompt);

// — Parse Response (expects Part 1 / Part 2 headers) —
const part1Match = aiResponse.match(/\[PART 1 - GROUP REPORT\]([\s\S]*?)\[PART 2 - INDIVIDUAL REPORT\]/);
const part2Match = aiResponse.match(/\[PART 2 - INDIVIDUAL REPORT\]([\s\S]*)/);
const part1 = part1Match ? part1Match[1].trim() : "Group report unavailable.";
const part2 = part2Match ? part2Match[1].trim() : "Individual report unavailable.";

// — Send Email —
const subject = `Your AI Literacy Report – ${group}`;
const body = buildEmailBody(participantName, part1, part2);
GmailApp.sendEmail(email, subject, "", { htmlBody: body, name: FROM_NAME });

Logger.log(`Email sent to ${email}`);
}

// — PROMPT BUILDER —
function buildPrompt(p) {
  const qLabels = ["Q1 (Generative AI)", "Q2 (AI in BIM)", "Q3 (Prompt Eng.)",
    "Q4 (Generative Design)", "Q5 (Hallucination)", "Q6 (Ethics/Legal)",
    "Q7 (LLMs)", "Q8 (Spec Risk)", "Q9 (Image Gen Limits)"];
  const qDetail = p.qScores.map((s, i) => `${qLabels[i]}: ${s === 1 ? "✓ Correct" : "✗ Incorrect"}`);
  const tierDist = Object.entries(p.tierCounts).map(([t,c]) => `${t}: ${c}`).join(", ");

  return `
You are an expert AI literacy evaluator for architects. Evaluate the following survey

```

== GROUP DATA ==

Group Name: \${p.group}
Group Size (responses so far): \${p.groupSize}
Group Average Score: \${p.groupAvg.toFixed(1)} / 9
Tier Distribution: \${tierDist}

== PARTICIPANT DATA ==

Name: \${p.participantName}
Years in Practice: \${p.yearsInPractice}
Total Score: \${p.totalScore} / 9
Competency Tier: \${p.tier}
Q10 Self-Assessment: \${p.q10}
Question-by-question results:
\${qDetail}

== INSTRUCTIONS ==

Write two clearly separated sections:

[PART 1 - GROUP REPORT]

- Address the whole group by name (\${p.group})
- State the group's average score and overall competency tier
- Describe the distribution of tiers across the group
- Highlight 2 areas the group collectively did well on, and 2 areas to improve
- End with an encouraging statement for the group's learning journey
- Keep this to 150-200 words. Professional, warm, constructive tone.

[PART 2 - INDIVIDUAL REPORT]

- Address \${p.participantName} personally
- Acknowledge their score (\${p.totalScore}/9) and competency tier (\${p.tier})
- Highlight 2-3 specific questions they answered correctly – explain WHY those show s
- If they answered any incorrectly, gently explain the correct concept (no shaming)
- Recommend 1-2 specific, actionable next steps tailored to their tier and years in
- Reference their Q10 self-assessment response: "\${p.q10}" – use this to personalise
- End with a motivating closing sentence
- Keep this to 200-250 words. Encouraging, specific, professional tone.

`;
}

// — OPENAI API CALL —————

```
function callOpenAI(prompt) {  
  const url = "https://api.openai.com/v1/chat/completions";  
  const payload = {  
    model: "gpt-4o",  
    messages: [{ role: "user", content: prompt }],  
    temperature: 0.7,  
    max_tokens: 800  
  };  
  const options = {  
    method: "post",  
    contentType: "application/json",  
    headers: { "Authorization": "Bearer " + OPENAI_API_KEY },  
    payload: JSON.stringify(payload),  
    muteHttpExceptions: true  
  };  
  const response = UrlFetchApp.fetch(url, options);  
  const json = JSON.parse(response.getContentText());
```

```

    return json.choices.message.content;
}

// — EMAIL HTML BUILDER —————
function buildEmailBody(name, part1, part2) {
    return `
<html><body style="font-family:Arial,sans-serif;max-width:680px;margin:auto;color:#
<div style="background:#1a3c5e;padding:24px;text-align:center;">
  <h2 style="color:#fff;margin:0;">SIA AI Literacy Assessment</h2>
  <p style="color:#aed4f5;margin:4px 0;">Your Personalised Competency Report</p>
</div>
<div style="padding:28px;">
  <p>Dear <strong>${name}</strong>,</p>
  <p>Thank you for completing the AI Literacy Survey. Below is your group report .
  <hr style="border:none;border-top:1px solid #ddd;margin:20px 0;">
  <h3 style="color:#1a3c5e;">☒ Part 1 – Group Report</h3>
  <p style="background:#f0f6ff;padding:16px;border-radius:8px;line-height:1.7;">§
  <hr style="border:none;border-top:1px solid #ddd;margin:20px 0;">
  <h3 style="color:#1a3c5e;">☒ Part 2 – Your Individual Assessment</h3>
  <p style="background:#f9f9f9;padding:16px;border-radius:8px;line-height:1.7;">§
  <hr style="border:none;border-top:1px solid #ddd;margin:20px 0;">
  <p style="font-size:12px;color:#999;">This report was generated automatically by
  For queries, contact your training coordinator.</p>
</div>
</html>`;
}

```

5.3 Setting Up the Trigger

1. In Google Apps Script editor, click Triggers (alarm icon, left sidebar)
2. Click + Add Trigger
3. Settings:
 - Function: `onFormSubmit`
 - Event source: **From spreadsheet**
 - Event type: **On form submit**
4. Save → Authorise when prompted (uses your Google account's Gmail to send)

Section 6: AI Evaluation Logic

6.1 What the AI Evaluates

The AI (GPT-4o or Gemini1.5 Pro) receives a structured prompt containing:[^1]

- **Group-level data:** group name, total respondents, average score, tier distribution
- **Individual data:** name, years in practice, score, tier, per-question right/wrong detail, Q10 self-assessment response

6.2 Evaluation Report Structure

Part 1 — Group Report (common to all in the same group)

☒ GROUP REPORT – [Group Name]

Group size: N participants

Average score: X.X / 9

Overall tier: [Tier Name]

Tier distribution:

- Foundational Awareness: N
- Applied Practitioner: N
- Strategic & Critical: N
- AI Champion: N

Collective strengths: [2 areas]

Areas for development: [2 areas]

Encouragement: [closing statement]

Part 2 — Individual Report (unique per participant)

☒ INDIVIDUAL REPORT – [Participant Name]

Score: X / 9 | Tier: [Tier Name]

Years in Practice: [X]

Strengths demonstrated:

- [Q answered correctly – why it matters]
- [Q answered correctly – why it matters]

Areas to develop:

- [Gently explains missed concept if any]

Recommended next steps:

1. [Specific action – tailored to tier + experience]
2. [Specific resource / learning path]

Personalised note: [references Q10 self-assessment]

[Motivating closing sentence]

6.3 Tone Calibration by Tier

Tier	AI Tone	Emphasis
Foundational	Welcoming, no-jargon	Demystify AI, safe exploration
Applied Practitioner	Encouraging, practical	Specific tool recommendations
Strategic & Critical	Peer-level, analytical	Ethics, leadership, upskilling others
AI Champion	Celebratory, aspirational	Community, mentoring, cutting-edge

Section 7: Sample AI-Generated Output

Sample Part 1 — Group Report

Group: ABC Architecture Pte Ltd — 12 participants completed the survey with an average score of 6.2 / 9, placing the group in the **Applied Practitioner** tier. The majority (7 participants) sit in the Applied tier, with 3 in Foundational and 2 reaching Strategic & Critical level.

The group demonstrated strong collective understanding of AI ethics and professional responsibility (Q6 — 11/12 correct) and AI hallucination risks (Q5 — 10/12 correct), which is commendable given the professional stakes involved. Areas for collective development include **technical understanding of LLMs** (Q7 — 6/12 correct) and **AI image generation limitations** (Q9 — 5/12 correct).

This is an excellent foundation. The diverse range of proficiency levels within your team is an asset — your AI Champions can become internal mentors as your practice deepens its AI capability. ☑

Sample Part 2 — Individual Report

Hi Sarah! Your score of 7 / 9 places you in the **Strategic & Critical** tier — the top 20% of participants. With 12 years of practice, you bring exactly the professional judgment needed to lead AI adoption thoughtfully.

You excelled particularly on **prompt engineering** (Q3) and **generative design** (Q4) — this shows you're already thinking beyond passive AI use toward actively directing AI tools as creative collaborators. Your correct answer on **specification writing risks** (Q8) reflects the kind of critical discernment that separates safe AI practitioners from reckless ones.

Your only missed questions were Q7 (how LLMs work technically) and Q9 (image generation limitations). These are niche technical areas — exploring a short course on *"AI Foundations for Design Professionals"* on Coursera would fill this gap in under 3 hours.

You mentioned you *"regularly integrate AI tools into daily practice"* — that's outstanding. Your next frontier is **sharing that knowledge**: consider proposing an internal lunch-and-learn at your firm. Your experience is exactly what your colleagues need to hear. You're on the right side of this shift. Keep going! ☑

Section 8: Alternative AI Setup (No-Code Option)

If direct API coding is not preferred, the following no-code alternatives can replace the Apps Script AI call:[^1]

Tool	Method	Cost
Zapier + OpenAI	Zap: Google Sheets → OpenAI Action → Gmail	~USD 20–50/mo
Make.com (Integromat)	Scenario: Sheets row → HTTP module (OpenAI) → Gmail	~USD 9–29/mo

Tool	Method	Cost
n8n (self-hosted)	Workflow: Sheets webhook → OpenAI node → Email	Free (self-hosted)
Gemini in Google Workspace	Experimental: Gemini Add-on for Sheets	Included in Workspace

For maximum control, reliability, and cost-efficiency at survey scale, **Apps Script + OpenAI GPT-4o** is the recommended approach.

Section 9: Data & Privacy Checklist

Item	Recommendation
Email data	Stored only in Google Drive — restrict sheet sharing to admins only
Mobile numbers	Optional field — do not include in AI prompt payload
AI API data	Use OpenAI's zero data retention API tier for PDPA compliance
Participant consent	Add a consent checkbox in the Google Form before Q1
Sheet access	Share "Scoring" sheet only with survey administrators
API key security	Store in Apps Script Properties Service (not hardcoded) — use <code>PropertiesService.getScriptProperties().setProperty("OPENAI_KEY", "sk-...")</code>

Section 10: Deployment Checklist

- ☐ Google Form created with all fields in correct order
- ☐ Form linked to Google Sheets (Responses → Create Spreadsheet)
- ☐ "Scoring" tab created with IF-formula scoring columns
- ☐ Apps Script code pasted and saved
- ☐ OpenAI API key added to Script Properties
- ☐ `onFormSubmit` trigger set (spreadsheet → on form submit)
- ☐ Test submission sent and email verified
- ☐ Consent checkbox added to Form
- ☐ QR code generated and tested on mobile
- ☐ Form confirmation message customised
- ☐ Sheet sharing restricted to admins

Section 11: Scaling & Enhancements

Enhancement	How
Multiple groups / events	Add a Group dropdown in Form; script groups by that field automatically
PDF report instead of email	Use Apps Script + DocumentApp or a PDF API to generate a report PDF as attachment
Dashboard for organisers	Add a "Dashboard" sheet tab with COUNTIF/AVERAGEIF formulas + embedded Charts
Delayed group report	Send Part 1 only after all expected group members submit (add a headcount field)
Certificate of participation	Trigger Google Slides template auto-fill for each participant
Reminder emails	Schedule a follow-up Apps Script time-trigger for non-responders via a separate tracking sheet

References

1. [AI_LiteracySurvey_MCQs.pdf](#) - Survey to assess AI literacy of respondents in SIA for planning of AI course for Architects

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